Chapter 11 — Hydrocarbons

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 261–264

Q.1 — Multiple Choice Questions (with correct option and reason)

I. An alkene undergoes ozonolysis followed by Zn/H_2O to give propanone and methanal. The alkene is:

(a) 1-Butene **(b)** 2-Butene

(c) 2-Methylpropene **(d)** 2-Methyl-2-butene

Answer: (c) 2-Methylpropene — $(CH_3)_2C{=}CH_2$ cleaves to $(CH_3)_2C{=}O$ (propanone) and HCHO (methanal).

II. Which reagent is typically used for acid-catalyzed hydration of alkenes to form alcohols?

(a) H_2/Ni **(b)** O_3 followed by Zn/H_2O

(c) Dilute H_2SO_4 **(d)** Br_2 in CCl_4

Answer: (c) Dilute H_2SO_4 — Dilute H_2SO_4 catalyses addition of water across the double bond (Markovnikov).

III. Halogenation of alkanes is an example of:

(a) Electrophilic substitution **(b)** Nucleophilic substitution

(c) Free-radical substitution **(d)** Oxidation

Answer: (c) Free-radical substitution — Alkanes react with halogens via a free-radical chain mechanism.

IV. Which reaction can an alkane undergo?

(a) Addition **(b)** Substitution

(c) Polymerization **(d)** Nitration

Answer: (b) Substitution — Alkanes undergo substitution (e.g. free-radical halogenation).

V. What is the first step in the electrophilic addition reaction of alkenes?

(a) Formation of a carbocation **(b)** Attack by a nucleophile

(c) Attack by an electrophile on the double bond **(d)** Formation of a free radical

Answer: (c) Attack by an electrophile on the double bond — The electron-rich double bond is first attacked by an electrophile.

VI. Addition of an unsymmetrical reagent to an unsymmetrical alkene follows:

(a) Markovnikov’s rule **(b)** Hund’s rule

(c) Le Chatelier’s principle **(d)** Aufbau principle

Answer: (a) Markovnikov’s rule — Markovnikov’s rule predicts the major product of such additions.

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VII. The most stable carbonium ion among the following is:

(a) CH_3^+ **(b)** $CH_3CH_2^+$

(c) $(CH_3)_2CH^+$ **(d)** $(CH_3)_3C^+$

Answer: (d) $(CH_3)_3C^+$ — Tertiary carbocations are most stabilised by the +I effect and hyperconjugation of three alkyl groups.

VIII. Markownikov's rule is applicable to:

(a) $CH_2=CH_2$ **(b)** $CH_3—CH_2—CH_3$

(c) $CH_3—CH=CH—CH_3$ **(d)** $(CH_3)_2C=CH_2$

Answer: (d) $(CH_3)_2C=CH_2$ — It applies to unsymmetrical alkenes; $(CH_3)_2C=CH_2$ is unsymmetrical.

IX. What intermediate is formed during electrophilic addition of HBr to an alkene?

(a) Carbocation **(b)** Carbanion

(c) Radical **(d)** Epoxide

Answer: (a) Carbocation — Protonation of the double bond gives a carbocation intermediate.

X. The enhanced stability of conjugated dienes compared to isolated dienes is primarily due to:

(a) Inductive effects of the double bonds. **(b)** Increased s-character of the hybridised orbitals.

(c) Delocalisation of π electrons across the conjugated system. **(d)** Steric hindrance between the double bonds.

Answer: (c) Delocalisation of π electrons across the conjugated system. — Delocalisation of π electrons lowers the energy of conjugated dienes.

XI. Which carbocation would be the least stable?

(a) $(CH_3)_2CH^+$ **(b)** $CH_3CH_2^+$

(c) CH_3^+ **(d)** $(CH_3)_3C^+$

Answer: (c) CH_3^+ — CH_3^+ (methyl cation) has no alkyl stabilisation and is least stable.

XII. Which is the repeating unit in poly(ethene)?

(a) $–CH_2CH_3–$ **(b)** –CHCH–

(c) $–CH_2CH_2–$ **(d)** $–CH(CH_3)CH_2–$

Answer: (c) $–CH_2CH_2–$ — Poly(ethene) is $–[CH_2–CH_2]_n–$.

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Q.2 — Short Answer Questions

a. Define: (i) Cycloalkanes (ii) Isomerism (iii) Conjugated dienes (iv) Inductive effect.

(i) Cycloalkanes: saturated cyclic hydrocarbons with general formula C_nH_{2n} (e.g. cyclohexane). (ii) Isomerism: compounds with the same molecular formula but different structures/arrangements. (iii) Conjugated dienes: dienes with alternating double and single bonds (C=C–C=C). (iv) Inductive effect: permanent displacement of σ electrons along a chain due to electronegativity difference.

b. Differentiate between: (i) Aliphatic and aromatic hydrocarbons (ii) Homolytic and heterolytic fission (iii) Electrophile and nucleophile.

(i) Aliphatic: open-chain or non-aromatic cyclic hydrocarbons; aromatic: contain a benzene-like delocalised π ring. (ii) Homolytic fission gives two radicals (each atom keeps one electron); heterolytic fission gives ions (one atom keeps both electrons). (iii) Electrophile: electron-pair acceptor (electron-loving); nucleophile: electron-pair donor (nucleus-loving).

c. Explain why alkanes do not undergo addition reactions.

Alkanes are saturated (only C–C and C–H single bonds) and have no π bond to break for addition. They undergo substitution instead.

d. How is an elimination reaction considered the opposite of an addition reaction?

Addition adds atoms across a double/triple bond to form a saturated product; elimination removes atoms from adjacent carbons to create a double/triple bond. They are reverse processes.

e. Compare the carbocation stability in propene and 2-butene.

Protonation of propene gives a secondary carbocation ($CH_3CH^+CH_3$). Protonation of 2-butene also gives a secondary carbocation ($CH_3CH^+CH_2CH_3$). Both are secondary; the 2-butene cation is slightly more stabilised by the larger alkyl group, but both are far more stable than a primary cation.

f. Given the molecular formula C_5H_{10}, list all possible structural isomers that are alkenes.

Pent-1-ene, pent-2-ene, 2-methylbut-1-ene, 3-methylbut-1-ene, 2-methylbut-2-ene. (Cycloalkanes also have formula C_nH_{2n} but are not alkenes.)

g. When propene undergoes electrophilic addition with HBr, it forms 2-bromopropane as the major product. Explain using carbocation stability.

H^+ adds to the terminal CH_2 to give the more stable secondary carbocation ($CH_3CH^+CH_3$) rather than a primary carbocation. Br^- then attacks the secondary cation, giving 2-bromopropane (Markovnikov's rule).

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h. Explain why conjugated alkenes may show different reactivity compared to isolated alkenes.

Conjugated dienes have delocalised π electrons, giving extra stability and allowing 1,2⁻ and 1,4-addition. Isolated dienes behave like independent alkenes and do not show this delocalisation.

i. Explain how inductive effects from alkyl groups stabilize carbocations in alkenes.

Alkyl groups release electron density through the +I effect, dispersing the positive charge on the carbocation. More alkyl groups (tertiary > secondary > primary > methyl) give greater stabilisation.

j. Write the equation for each reaction: (i) $CH_3CH_2CH=CH_2 + H_2$ (Ni) (ii) $CH_3CH=CH_2 + Cl_2$ (iii) $CH_3CH_2CH=CHCH_2CH_3 + H_2O$ (H_2SO_4).

(i) $CH_3CH_2CH=CH_2 + H_2 \rightarrow CH_3CH_2CH_2CH_3$ (butane).

(ii) $CH_3CH=CH_2 + Cl_2 \rightarrow CH_3CHClCH_2Cl$ (1,2-dichloropropane).

(iii) $CH_3CH_2CH=CHCH_2CH_3 + H_2O \rightarrow CH_3CH_2CH_2CH(OH)CH_2CH_3$ (hexan-3-ol; the symmetrical alkene gives a single alcohol).

k. Write structural formulas for: (i) Isobutylene (ii) 2,3,4,4-tetramethyl-2-pentene (iii) 2,5-heptadiene (iv) 4,5-dimethyl-2-hexene (v) Vinylacetylene (vi) 1,3-pentadiene (vii) 1-butyne (viii) 3-n-propyl-1,4-pentadiene (ix) Vinyl bromide (x) But-1-en-3-yne (xi) 4-methyl-2-pentyne (xii) Isopentane.

(i) $(CH_3)_2C=CH_2$. (ii) $CH_3C(CH)=C(CH_3)C(CH_3)_2CH_3$. (iii) $CH_3CH=CHCH_2CH=CHCH_3$. (iv) $CH_3CH=CHCH(CH_3)CH(CH_3)CH$.

(v) $CH_2=CH–C\equiv CH$. (vi) $CH_2=CH–CH=CH–CH_3$. (vii) $CH\equiv C–CH_2–CH_3$. (viii) $CH_2=CH–CH(CH_2CH_2CH_3)–CH=CH_2$.

(ix) $CH_2=CHBr$. (x) $CH_2=CH–C\equiv CH$. (xi) $CH_3C\equiv CCH(CH_3)CH_3$. (xii) $(CH_3)_2CHCH_2CH_3$.

l. Write IUPAC names of the given compounds.

Names (interpreting the printed condensed structures):

Item	IUPAC name
i	Hex-2-ene
ii	2-Methylpropene (assuming $(CH_3)_2C=CH_2$)
iii	2⁻(Propan-2-yl)pent-1-ene
iv	Buta-1,3-diene
v	2-Ethylpent-1-ene
vi	3-Ethylpentane
vii	3,3,4-Trimethylhexane

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Item	IUPAC name
viii	Hexa-1,3,5-triene
ix	2,2,4,4-Tetramethylpentane (assuming $(CH_3)_3CCH_2C(CH_3)_3$)
x	But-1-en-3-yne (vinylacetylene)
xi	2,2-Dimethylbutane
a	3-Ethylhexane
b	2,3,4-Trimethylpentane
c	2,4-Dimethylpentane (interpreted)

Textbook check (p. 264): several condensed structures omit subscripts/bonds (ii, ix, x, c). The names above use the chemically valid structures intended; the printed formulas are incomplete.

Descriptive / Long Questions

Q.3 Describe the free radical halogenation of methane using Cl_2 as an example.

The reaction proceeds by a free-radical chain mechanism under UV light/heat.

Initiation: $Cl_2 \rightarrow 2Cl\bullet$ (homolytic fission).

Propagation: $Cl\bullet + CH_4 \rightarrow HCl + \bullet CH_3$; $\bullet CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\bullet$.

Termination: $Cl\bullet + Cl\bullet \rightarrow Cl_2$; $\bullet CH_3 + Cl\bullet \rightarrow CH_3Cl$; $\bullet CH_3 + \bullet CH_3 \rightarrow C_2H_6$.

Further substitution can give CH_2Cl_2, $CHCl_3$ and CCl_4.

Q.4 Describe the following methods for the preparation of alkenes: (i) Dehydrohalogenation of alkyl halides (ii) Dehydration of alcohols.

(i) Dehydrohalogenation: heating an alkyl halide with alcoholic KOH removes HX to form an alkene. Example: $CH_3CH_2Br + KOH(alc) \rightarrow CH_2{=}CH_2 + KBr + H_2O$. The more substituted alkene is usually major (Saytzeff).

(ii) Dehydration: heating an alcohol with concentrated H_2SO_4 (or Al_2O_3) removes water to form an alkene. Example: $CH_3CH_2OH \rightarrow$(conc. H_2SO_4, 170 °C) $CH_2{=}CH_2 + H_2O$.

Q.5 Describe the mechanism of electrophilic addition of hydrogen halides to alkenes. Discuss Markovnikov's rule.

Step 1: the π electrons of the alkene attack the H of HX, forming a C–H bond and a carbocation intermediate (the more stable carbocation forms preferentially). Step 2: the halide ion X^- attacks the carbocation to give the alkyl halide.

Markovnikov's rule: when an unsymmetrical reagent (HX) adds to an unsymmetrical alkene, the H attaches to the carbon with more hydrogens and X to the carbon with fewer hydrogens, because this proceeds via the more stable

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carbocation. Example: $CH_3CH=CH_2 + HBr \rightarrow CH_3CHBrCH_3$ (2-bromopropane).

Q.6 Explain the following reactions of alkenes with examples: (a) Halogenation (b) Ozonolysis (c) Epoxidation (d) Polymerization.

(a) Halogenation: addition of X_2 across the double bond. $CH_2=CH_2 + Br_2 \rightarrow CH_2BrCH_2Br$ (decolourises bromine water).

(b) Ozonolysis: reaction with O_3 followed by Zn/H_2O cleaves the double bond to carbonyl compounds. $CH_3CH=CH_2 \rightarrow CH_3CHO + HCHO$.

(c) Epoxidation: reaction with a peracid (e.g. RCO_3H) gives an epoxide (oxirane). $CH_2=CH_2 + RCO_3H \rightarrow$ ethylene oxide + RCO_2H.

(d) Polymerization: many alkene monomers join to form a polymer. $nCH_2=CH_2 \rightarrow -[CH_2–CH_2]_n–$ (polyethene).

Quick Check Solutions (in-chapter practice boxes)

Quick Check 11.1 (book p. 241)

a. Structural formulas for 2-methylbutane and 2,2-dimethylpropane.

2-Methylbutane: $(CH_3)_2CHCH_2CH_3$. 2,2-Dimethylpropane: $C(CH_3)_4$.

b. Displayed formula of 2,3,3-trimethylhexane and methylcyclohexane.

2,3,3-Trimethylhexane: $CH_3CH(CH_3)C(CH_3)_2CH_2CH_2CH_3$. Methylcyclohexane: a six-membered carbon ring with one $–CH_3$ substituent.

c. Two differences between cyclopentane and pentane.

Cyclopentane is cyclic (C_5H_{10}) while pentane is open-chain (C_5H_{12}); cyclopentane has two fewer H atoms and a ring structure, giving slightly different physical properties.

d. Molecular formula of eicosane (20-carbon straight-chain alkane).

$C_{20}H_{42}$ (alkane general formula C_nH_{2+2}).

Quick Check 11.2 (book p. 243)

a. Why branched alkanes have lower boiling points than straight-chain alkanes.

Branching makes molecules more compact, reducing surface area and London dispersion forces, so less energy is needed to separate them.

b. Why alkanes have a tetrahedral shape.

Each carbon is sp^3 hybridised with four bond pairs arranged tetrahedrally at ~109.5° to minimise repulsion.

c. Draw the shapes of cyclopropane and cyclopentane.

Cyclopropane: a triangular ring of three CH_2 groups. Cyclopentane: a pentagonal (puckered) ring of five CH_2 groups.

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Quick Check 11.3 (book p. 246)

a. Why alkanes have high stability.

They contain strong, non-polar C–C and C–H σ bonds and have no reactive π bonds or functional groups.

b. Major types of reactions alkanes undergo.

Substitution (e.g. free-radical halogenation) and combustion (oxidation).

c. How the termination step occurs in halogenation of alkanes.

Two radicals combine to form a stable molecule, e.g. $Cl\bullet + Cl\bullet \rightarrow Cl_2$; $\bullet CH_3 + Cl\bullet \rightarrow CH_3Cl$; $\bullet CH_3 + \bullet CH_3 \rightarrow C_2H_6$.

d. Conditions under which a mixture of halogenoalkanes is obtained.

Excess halogen and continued UV light/heat allow successive substitution, giving mono⁻, di⁻, tri⁻ and tetra-halogenated products.

e. Products of free-radical chlorination of ethane.

Chloroethane (CH_3CH_2Cl), then 1,1⁻ and 1,2-dichloroethane, and further polychlorinated products up to hexachloroethane.

Quick Check 11.4 (book p. 248)

a. Reactions of propanol with H_2SO_4.

With conc. H_2SO_4 at ~170 °C, dehydration gives propene: $CH_3CH_2CH_2OH \rightarrow CH_3CH{=}CH_2 + H_2O$. At lower temperature (~140 °C) intermolecular dehydration can give dipropyl ether.

b. Name the alkenes: (i) $CH_3C(CH_3){=}CHCH_3$ (ii) $CH_2{=}CHCH(CH_3)_2$.

(i) 2-Methylbut-2-ene. (ii) 3-Methylbut-1-ene.

c. Structural formulas: (i) 3-methyl-2-butene (ii) 2-methyl-4-ethyl-2-pentene.

(i) $CH_3CH{=}C(CH_3)CH_3$. (ii) $CH_3C(CH_3){=}CHCH(C_2H_5)CH_3$.

d. Dehydrohalogenation of bromopropane.

$CH_3CH_2CH_2Br + KOH(alc) \rightarrow CH_3CH{=}CH_2 + KBr + H_2O$.

e. Can propanol undergo dehydration? Give reactions.

Yes. $CH_3CH_2CH_2OH \rightarrow$(conc. H_2SO_4, heat) $CH_3CH{=}CH_2 + H_2O$ (propene).

Quick Check 11.5 (book p. 250)

a. How an alkane differs from an alkene in stability.

Alkanes are more stable (only strong σ bonds); alkenes have a reactive π bond that is more easily attacked, making them less stable/more reactive.

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b. Which species is likely to act as a nucleophile: H_2, H^+, OH^-? Why?

OH^-, because it is electron-rich (negative charge/lone pairs) and donates an electron pair.

c. Which species is likely to act as an electrophile: H_2, H^+, OH? Why?

H^+, because it is electron-deficient and accepts an electron pair.

Quick Check 11.6 (book p. 252)

a. How alkenes react with an electrophile.

The electron-rich π bond attacks the electrophile, forming a carbocation intermediate; a nucleophile then attacks the carbocation to give the addition product.

b. Why carbocation stability is 3° > 2° > 1°.

More alkyl groups stabilise the positive charge by the +I (inductive) effect and hyperconjugation; tertiary has three alkyl groups, secondary two, primary one.

Quick Check 11.7 (book p. 254)

a. Compare the relative rates of addition of HCl, HBr and HI to alkenes.

HI > HBr > HCl. The H–X bond strength decreases down the group, so HI adds most readily.

b. Difference between addition and elimination reactions.

Addition adds atoms across a double/triple bond to give a more saturated product; elimination removes atoms from adjacent carbons to create a double/triple bond.

c. How an alkene reacts with an electrophile.

The π electrons attack the electrophile, forming a carbocation, which is then captured by a nucleophile.

d. Apply Markovnikov's rule to addition of HBr to 2-pentene.

2-Pentene ($CH_3CH{=}CHCH_2CH_3$) is nearly symmetrical; HBr addition can give 2-bromopentane and 3-bromopentane. Markovnikov's rule places H on the carbon with more H atoms, but here both alkene carbons bear one H, so selectivity is small.

Quick Check 11.8 (book p. 256)

a. Equation for ozonolysis of ethene.

$CH_2{=}CH_2 + O_3 \rightarrow$ (ozonide) $\rightarrow$(Zn/H_2O) 2HCHO (methanal).

b. Epoxidation of 2-butene.

$CH_3CH{=}CHCH_3 + RCO_3H \rightarrow$ 2,3-dimethyloxirane (an epoxide) + RCO_2H.

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c. How ozonolysis indicates the position of a double bond.

The double bond is cleaved and each carbon becomes a carbonyl group. Identifying the carbonyl products reveals where the double bond was. Example: but-2-ene gives two ethanal molecules; but-1-ene gives methanal + propanal.

Quick Check 11.9 (book p. 257)

a(i). Why ethene is suitable for polymerization.

Its C=C double bond contains a reactive π bond that can open and link many monomers into a long chain (addition polymerization).

a(ii). Change in hybridization and bond angle during polymerization.

Carbon changes from sp^2 (trigonal planar, ~120°) in ethene to sp^3 (tetrahedral, ~109.5°) in polyethene.

b. Why an addition reaction increases saturation.

Addition converts a π bond into new σ bonds, reducing the number of multiple bonds and increasing the hydrogen content/saturation.

Quick Check 11.10 (book p. 257)

a. Draw the structure of penta-1,3-diene.

$CH_2=CH–CH=CH–CH_3$ (conjugated diene with alternating double and single bonds).

b. Illustrate delocalization of electrons in conjugated dienes.

The p orbitals on four adjacent carbons overlap, allowing the π electrons to spread over the whole C=C–C=C system rather than being localised in two separate double bonds.

Quick Check 11.11 (book p. 261)

a. Isomers of C_4H_8 showing positional isomerism.

But-1-ene ($CH_2=CHCH_2CH_3$) and but-2-ene ($CH_3CH=CHCH_3$) are positional isomers (double bond in different positions).

b. What type of isomers are ethoxy methane ($CH_3OC_2H_5$) and propanol (C_3H_7OH)?

Functional group isomers (ether vs alcohol).

c. Types of structural isomerism in alkenes.

Chain isomerism, position isomerism (of the double bond), and functional group isomerism (alkene vs cycloalkane).

d. Different isomers of $C_4H_9NH_2$.

Butan-1-amine, butan-2-amine, 2-methylpropan-1-amine, and 2-methylpropan-2-amine.

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Textbook checks flagged in this chapter

• p. 264, Q.2(l): several condensed structures omit subscripts or bonds (items ii, ix, x and c); the names use the intended valid structures.

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